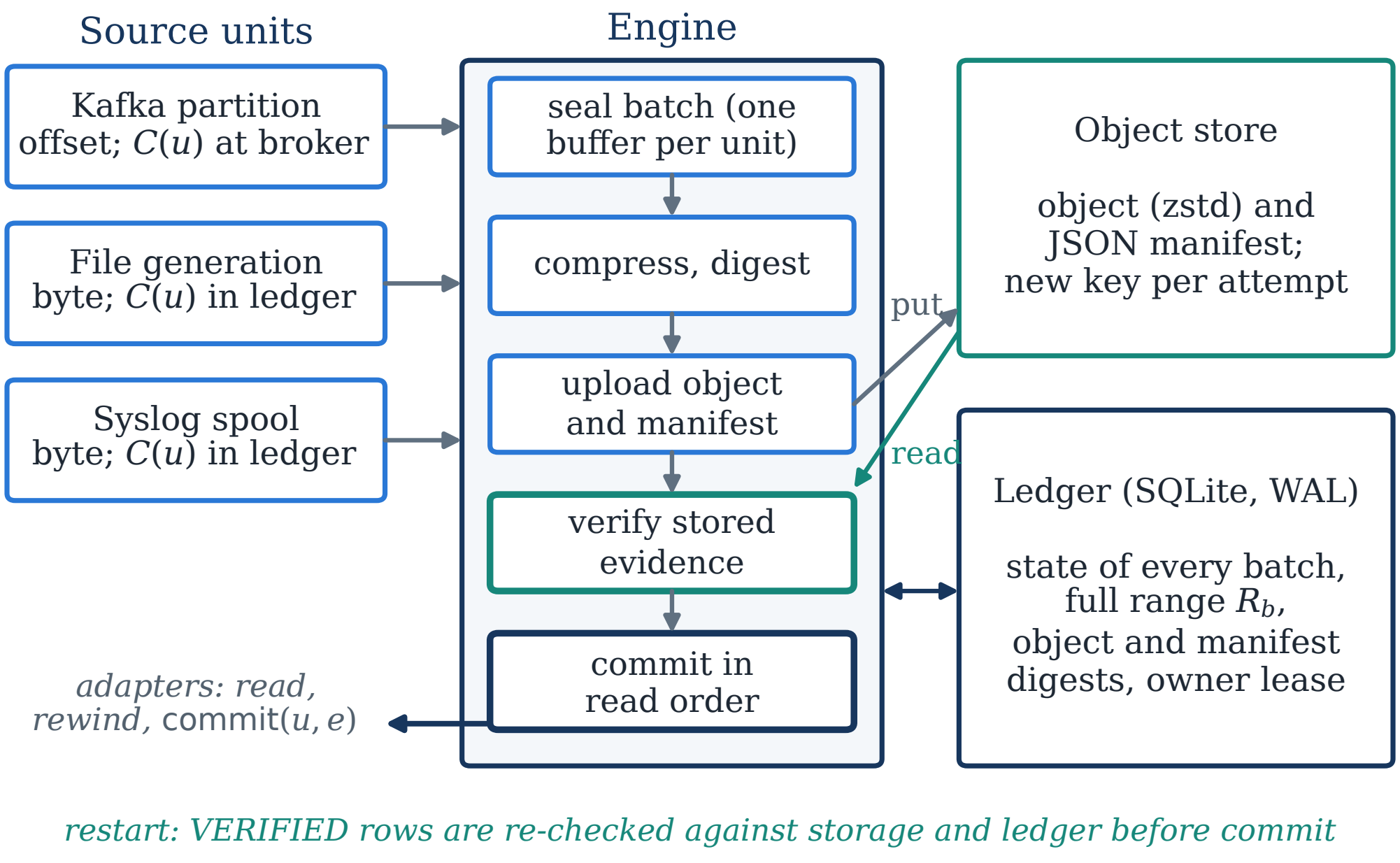
Source units
Engine
Kafka partition offset; $C(u)$ at broker
File generation byte; $C(u)$ in ledger
Syslog spool byte; $C(u)$ in ledger
seal batch (one buffer per unit)
compress, digest
upload object and manifest
verify stored evidence
commit in read order
put
read
Object store
object (zstd) and JSON manifest; new key per attempt
Ledger (SQLite, WAL)
state of every batch, full range R_b, object and manifest digests, owner lease
adapters: read, rewind, commit(u, e)
restart: VERIFIED rows are re-checked against storage and ledger before commit